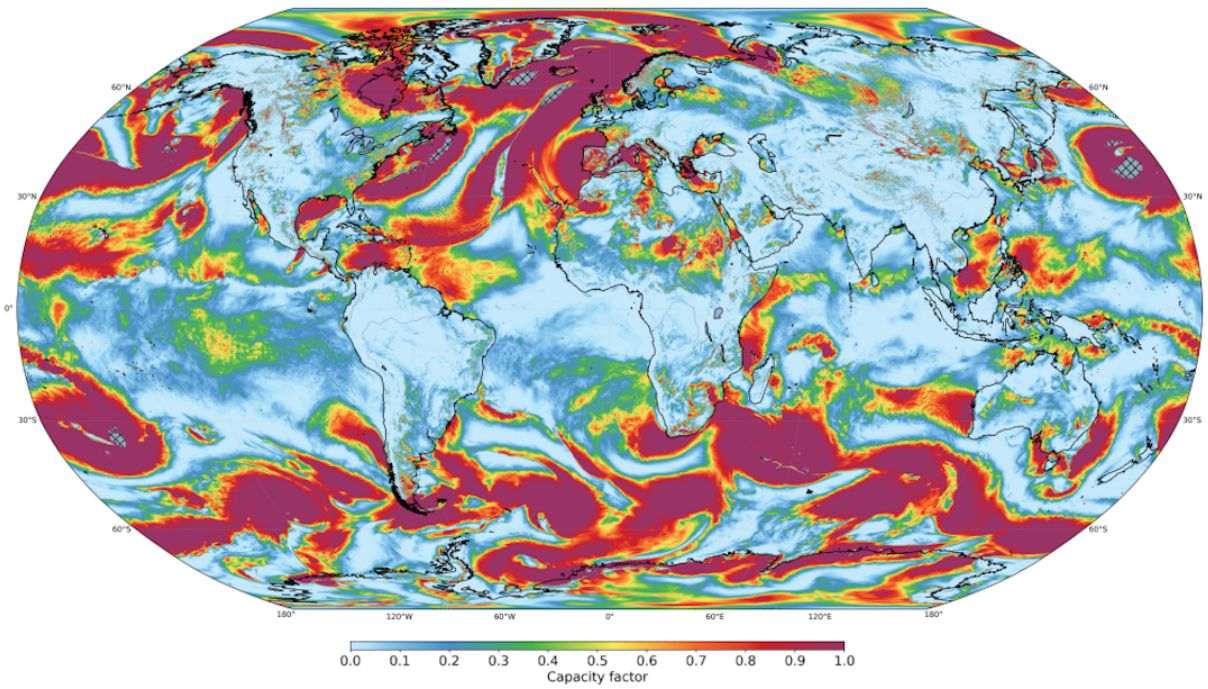


Figure 1.10 An example of a comprehensive digital twin—a view of indicators relevant to wind energy including hourly wind speed distribution with its changes at the multi decadal scale to help improve wind farm design generated with the Climate Change Adaptation Digital Twin, part of Destination Earth. Image © ECMWF. Licensed under CC BY 4.0.

1.3.5 Autonomous digital twin

An *autonomous digital twin* extends beyond purely modeling the physical system and closes the loop between the physical and digital realms by taking real world actions based on information, predictions, or simulations in the digital system. These actions can range from direct physical actuation—for example, opening a pressure release valve when tank pressure is predicted to exceed safe limits—to electronic actuation that ultimately results in an action such as the raising of a notification on a user's device.

Systems resembling autonomous digital twins have traditionally been found in industrial processes, but it's important to distinguish between different types of automated systems. Closed-loop process control systems, such as automated plant control (APC) systems managed by distributed control systems (DCS), use data from SCADA systems to control PLCs through pre-